

TASK 5 SUBMISSION

Competitor Demand Prediction Using Job Postings

Hiring Trend Analysis

Trend tables and visual summaries for NVIDIA

Prepared by	Nayab Khalid
Company specialism	NVIDIA
Task	Task 5 of 10 - Hiring Trend Analysis
Window	2026-05-13 to 2026-09-02, 12 observations over 112 days
Date	20 September 2026
Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

Measure	Value
Open roles	min 395, max 433, mean 416
Net change over the window	+18 roles (+4.6%)
New postings per day	mean 14.8, range 9.4 to 20.9
Closed postings per day	mean 14.7, range 10.4 to 20.3
Trend in new per day	+0.60 per observation
Median time a completed posting stayed open	14 days

The headline

NVIDIA's open-role count barely moved - 395 to 413, never leaving a band of 395 to 433. The flows underneath it did. Postings opened per day rose from about 12 to about 17, and closings rose in step. The stock is flat while the churn is accelerating: NVIDIA is not expanding its open-role list, it is turning that list over faster.

1 Method, and why it is not a plain weekly count

Three properties of the Task 2 data decide how this had to be computed.

1.1 The observation gaps are uneven

7, 21, 14, 7, 7, 21, 7, 7, 8, 7 and 6 days, because seven partial scrapes were excluded in Task 2. Counting arrivals per observation would turn the two 21-day gaps into fake hiring spikes: they show **266 and 297 arrivals** against a typical 100 to 120. Normalised per day they are **12.7 and 14.1**, sitting exactly among their neighbours. Every flow in this task is therefore a per-day rate, with the raw counts kept alongside so the normalisation is visible rather than hidden.

1.2 The stock is observed; the flows are derived

Open roles on a date is a direct count from that day's snapshot. New and closed are set differences between consecutive snapshots, so they inherit whatever happened inside the gap. Where the two disagree, the stock is the more trustworthy series.

1.3 Both ends are censored

The 395 postings already open at the first observation are stamped with that date, so the first period's arrivals are not arrivals; that period is dropped from all flow analysis. At the other end, 413 postings were still open, so their lifespans are unknown rather than short.

This matches the team's shared config on weekly buckets and zero-filled empty periods, with the per-day normalisation added because our gaps are not uniform. That addition needs raising with the team, since any member with a gapped series has the same problem.

2 Open roles: flat

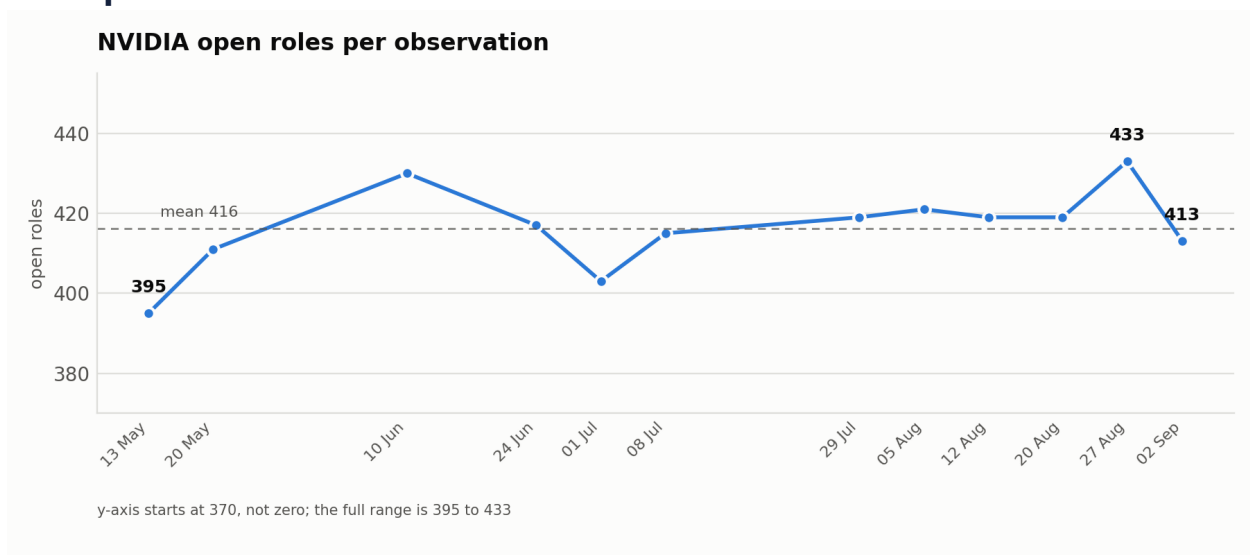


Figure 1. Open roles at each observation. Points are placed at their true dates, so the two 21-day gaps are visible as wider spacing.

The series sits in a 38-role band for four months. The only visible move is a rise to 433 on 27 August followed by a fall to 413 a week later, and at a mean of 416 that is a 5% wobble, not a trend. If the question is whether NVIDIA is expanding, the answer from this window is no: the open-role count is being held roughly constant.

3 Hiring velocity: accelerating

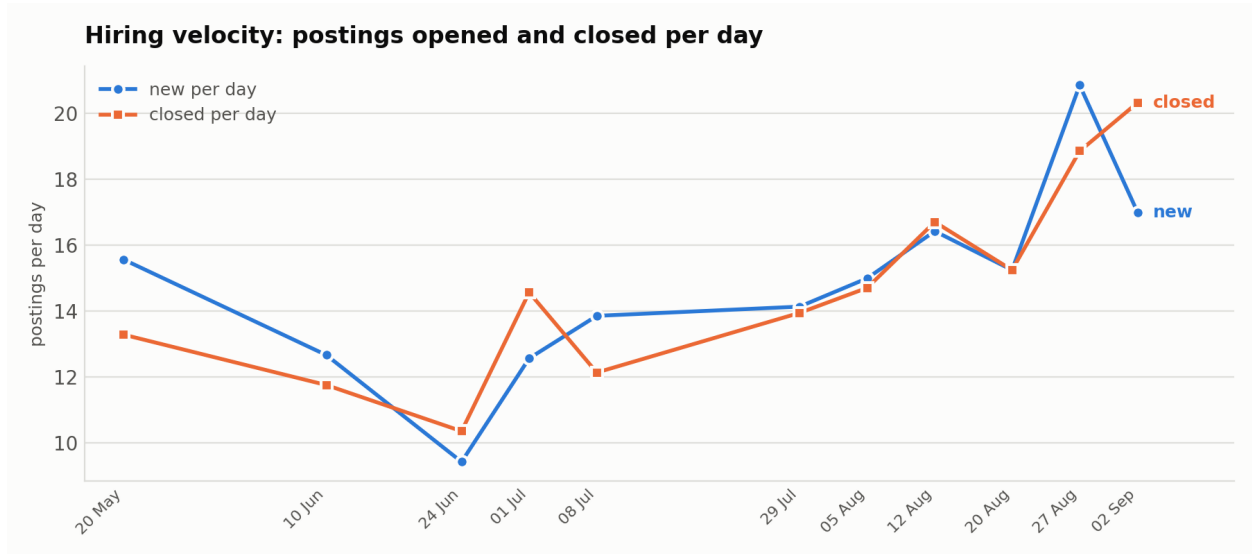


Figure 2. Postings opened and closed per day. The first period is excluded because it is left-censored.

Observation	Gap	New	Closed	New/day	Closed/day	Net/day
2026-05-20	7	109	93	15.57	13.29	+2.29
2026-06-10	21	266	247	12.67	11.76	+0.90
2026-06-24	14	132	145	9.43	10.36	-0.93
2026-07-01	7	88	102	12.57	14.57	-2.00
2026-07-08	7	97	85	13.86	12.14	+1.71
2026-07-29	21	297	293	14.14	13.95	+0.19
2026-08-05	7	105	103	15.00	14.71	+0.29
2026-08-12	7	115	117	16.43	16.71	-0.29
2026-08-20	8	122	122	15.25	15.25	0.00
2026-08-27	7	146	132	20.86	18.86	+2.00
2026-09-02	6	102	122	17.00	20.33	-3.33

The two lines track each other closely. Net change per day never exceeds 3.3 in either direction. Postings are replaced at almost exactly the rate they leave.

Both rates rise together. The three-observation rolling mean of new-per-day goes from 14.1 in June to 17.7 at the start of September, a rise of about 25%. Closings follow the same path. A company adding headcount would show new outrunning closed; these move in lockstep.

The trough is late June. New-per-day bottoms at 9.4 on 24 June, the only observation below 10, and it is the same period where net change is most negative. Whether that is a real summer pause or a collection artefact cannot be settled from one occurrence.

4 How long a posting stays open

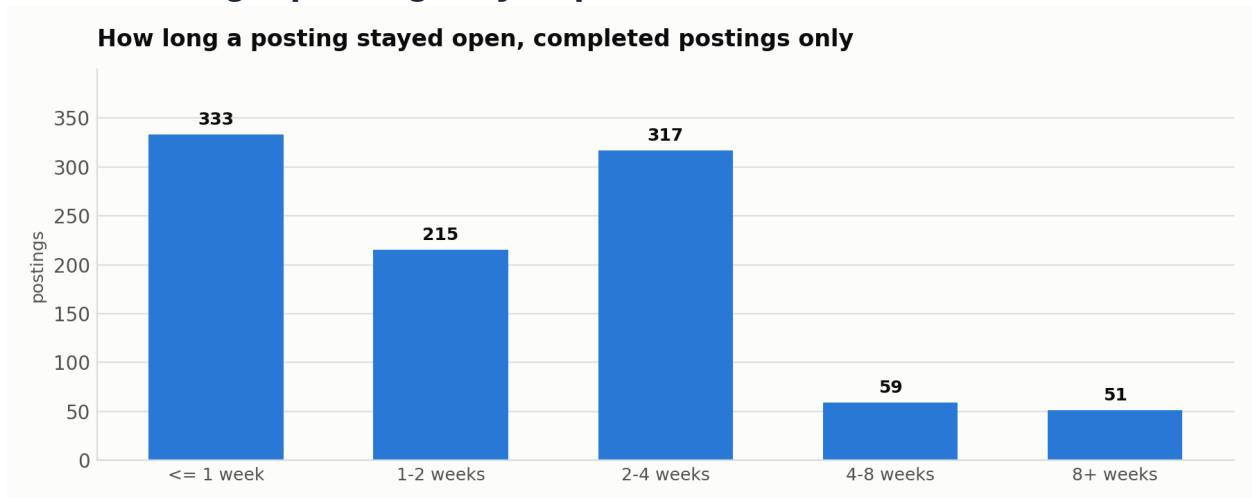


Figure 3. Lifespan of completed postings. The 413 still open on 2 September are excluded rather than counted as short-lived.

Bucket	Postings	Share
1 week or less	333	34.2%
1 to 2 weeks	215	22.1%
2 to 4 weeks	317	32.5%
4 to 8 weeks	59	6.1%
8 weeks or more	51	5.2%

Median 14 days across 975 completed postings; a third are gone within a week. **These values are quantised by the sampling cadence:** a posting observed once has a measured span of 0 days, twice a week apart has 7. The buckets are really "how many observations it survived", so the shape is trustworthy and the precise values are not.

Closed does not mean filled

A short median is consistent either with roles being filled quickly or with postings being relisted under new identifiers. Nothing in this data distinguishes a filled role from a cancelled or re-posted one, and no conclusion here assumes otherwise.

5 What is being hired for, and how that shifted

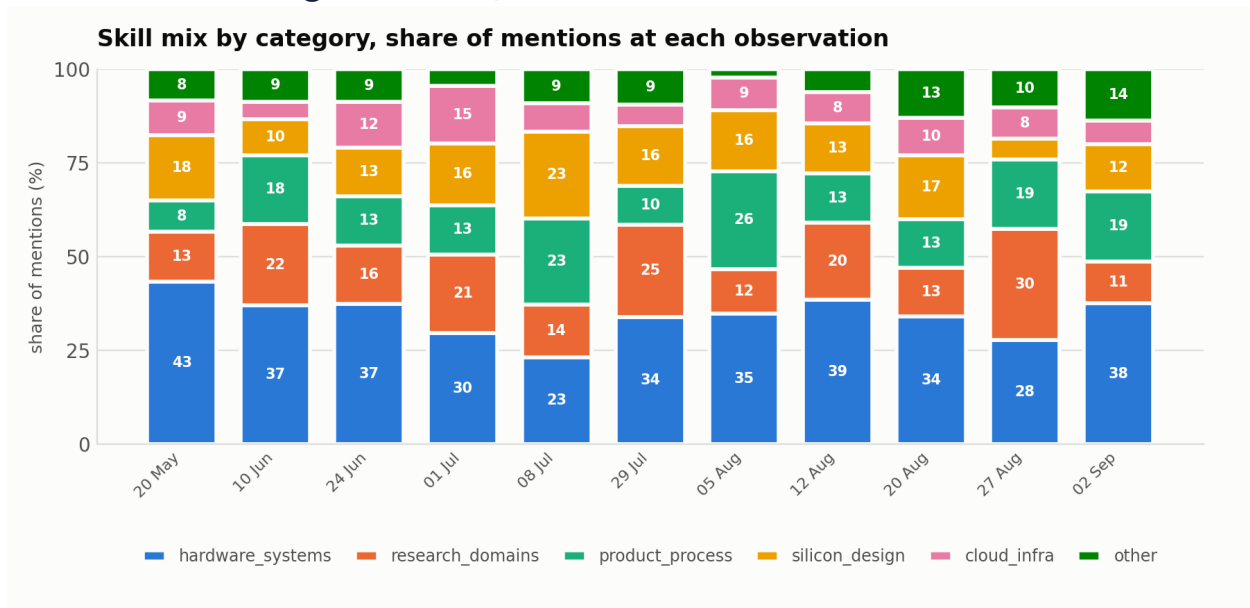


Figure 4. Skill mix as share of mentions. Plotted as share, not counts: a count chart would be dominated by the censored first observation and the two 21-day gaps.

Category	First observation	Last observation	Direction
hardware_systems	36.6%	37.5%	flat
product_process	13.7%	18.8%	up
research_domains	16.1%	11.3%	down
silicon_design	15.2%	12.5%	down
cloud_infra	8.9%	6.3%	down
other	9.4%	13.8%	up

Hardware and systems hold a steady third of all skill mentions throughout: that is NVIDIA's centre of gravity and it does not move. The visible rotation is out of research domains and into product and go-to-market roles, which sits alongside the Task 4 finding that Solutions Architecture is the single largest skill in the corpus at 192 postings.

6 Key patterns

1. **Flat stock, accelerating churn.** Open roles +4.6% over four months while both flow rates rose about 25%. NVIDIA is replacing roles faster, not adding them.
2. **Replacement, not expansion.** New and closed track within 3.3 postings per day of each other at every single observation.
3. **Hardware is the constant.** A steady 37% of skill mentions, unmoved across the window.
4. **The rotation is toward the field.** Product and go-to-market up 5 points, research down 5.
5. **Postings turn over quickly.** Median 14 days open, a third gone inside a week.

7 Limitations

#	Limitation
1	Eleven usable flow periods. After dropping the censored first observation the trend rests on 11 intervals with uneven gaps. It shows direction, not magnitude, and cannot support seasonality.
2	Four months is not a business cycle. The late-June trough cannot be separated from a seasonal effect with a single year of data.
3	Closed is not filled.
4	European bias: the Task 2 source curates EU-office employers, so the flat stock may not hold worldwide.
5	Lifespans are quantised to the observation cadence.
6	Skill mix inherits the Task 4 ceiling: titles carry about one skill each, so the mix is coarser than a description-based one would be.

8 For the team

1. **Agree per-day normalisation for gapped series.** Anyone whose collection missed days has the same distortion. Raw per-observation counts are not comparable across members with different cadences.
2. **Agree how to treat censored first observations,** so Task 6 does not compare my de-censored series against someone's raw one.
3. **Stock versus flow needs one definition.** I can report both. A member collecting only current openings has a stock series alone and no flows, and Task 6 must compare like with like.

Files

File	Contents
data/trends/nvidia_weekly_trend_20260902.csv	The main trend table: stock, flows, per-day rates, rolling mean
data/trends/nvidia_category_mix_20260902.csv	Skill mentions by category at each observation
data/trends/nvidia_lifespan_20260902.csv	Lifespan distribution for completed postings
figures/fig1 ... fig4 .png	The four figures above
analyse_trends.py	The analysis, re-runnable from a clean checkout

```
cd work/task-5/nvidia-nayab-khalid
python analyse_trends.py
```

Declaration	No personal or sensitive data is present in any output. No new data was collected for this task; the inputs are the corpus from Task 3 and the extracted skills from Task 4.
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Date	20 September 2026